

Paper 1B drill: reading graphs

Three graphs. Get the numbers off them.

IB Physics (2023 syllabus)

with worked answers

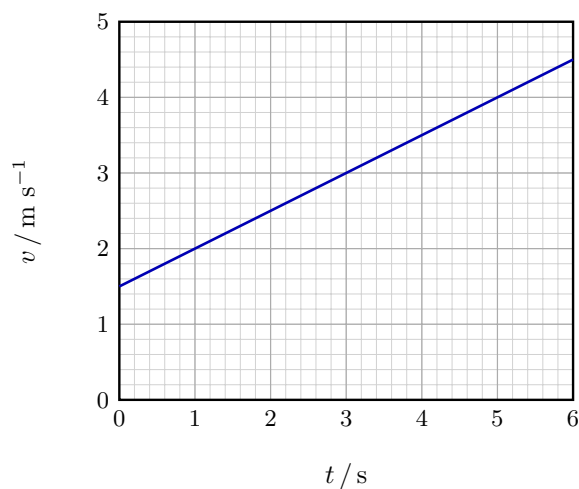
Name: _____

Date: _____

Total: **22 marks**

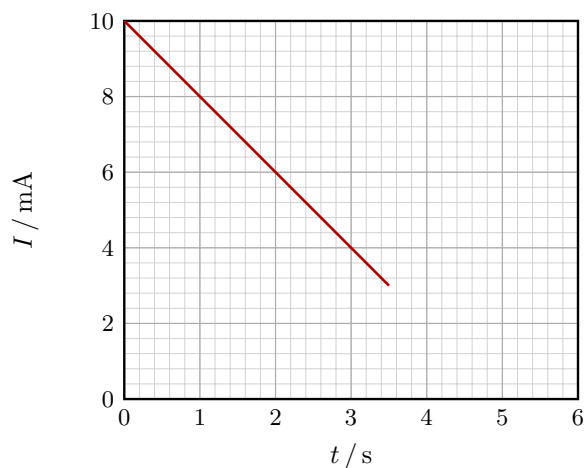
Take readings from the grid, not from the equation of the line; there isn't one. For a gradient, use two points far apart on the line, not two of the plotted data points.

Graph 1 *a trolley on a ramp*



1. Read the velocity at $t = 4.0 \text{ s}$. [1]
2. State the intercept on the velocity axis, and say what it means physically. [2]
3. Determine the gradient, with its unit, and say what it represents. [3]

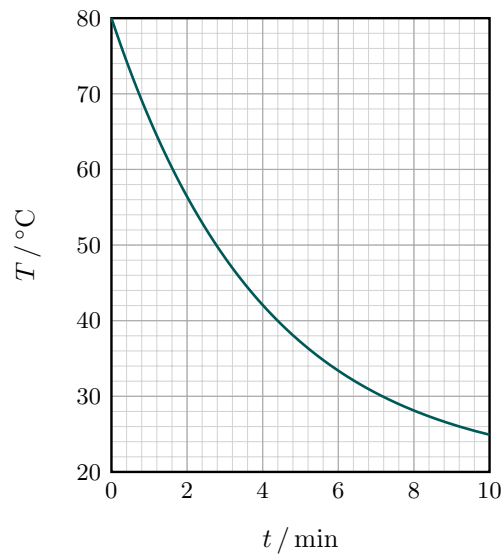
Graph 2 *a capacitor discharging through a fixed resistor*



4. Read the current at $t = 2.0 \text{ s}$. [1]

5. Determine the gradient, with its unit. [2]
6. The measurements stopped at $t = 3.5$ s. Estimate when the current would reach zero, and state what you had to do to the line to find out. [3]
7. Suggest why the real current would not in fact reach zero at that moment. [2]

Graph 3 *a beaker of water cooling in a room*



8. Read the temperature at $t = 2.0$ min. [1]
9. The curve flattens towards a value it never quite reaches. State that value and what it is physically. [2]
10. State where on the curve the water is cooling fastest, and explain why in terms of the graph and in terms of the physics. [3]
11. Explain why a gradient read at a single point on this graph cannot be found in the same way as for Graphs 1 and 2. [2]

Answers

Graph 1 *trolley on a ramp*

1. $v = 3.5 \text{ m s}^{-1}$.
2. The line meets the velocity axis at 1.5 m s^{-1} . This is the velocity at $t = 0$, so the trolley was already moving when timing began; it was given a push rather than released from rest.
3. Taking the two ends of the line, (0, 1.5) and (6.0, 4.5):

$$\text{gradient} = \frac{4.5 - 1.5}{6.0 - 0} = 0.50 \text{ m s}^{-2}.$$

The gradient of a velocity–time graph is the acceleration, so the trolley accelerates uniformly at 0.50 m s^{-2} .

Graph 2 *capacitor discharging*

4. $I = 6.0 \text{ mA}$.
5. Using (0, 10.0) and (3.5, 3.0):

$$\text{gradient} = \frac{3.0 - 10.0}{3.5 - 0} = -2.0 \text{ mA s}^{-1}.$$

The negative sign matters: the current is falling.

6. Extending the straight line beyond the last measurement, it crosses the time axis at $t = 5.0 \text{ s}$. This is *extrapolation*: reading a value from outside the range of the data by continuing the trend, as opposed to interpolation between measured points.
 7. Extrapolation assumes the trend continues, and here it does not. A capacitor discharges exponentially, so the graph is really a curve that flattens as the current falls and approaches zero asymptotically rather than crossing. Over the short measured range it merely *looks* straight. Any extrapolation is only as good as the assumption that the pattern holds outside the data, which is exactly what has not been tested here.
-

Graph 3 *water cooling*

8. About 56°C . Accept 55 to 57.
9. It approaches 20°C , which is room temperature. Once the water reaches it there is no temperature difference left to drive any further transfer of thermal energy.
10. At the very start, $t = 0$. On the graph, that is where the curve is steepest, and the gradient of a temperature–time graph is the rate of cooling. Physically, the rate of energy transfer depends on the temperature difference between the water and the room, and that difference is largest at the beginning, so the cooling is fastest then and slows continuously as the gap closes.
11. Graphs 1 and 2 are straight, so their gradient is the same everywhere and any two points on the line give it. This graph is curved, so the gradient changes from point to point: there is no single value. To find the rate of cooling at one instant you would have to draw a tangent to the curve at that point and take the gradient of the tangent instead.